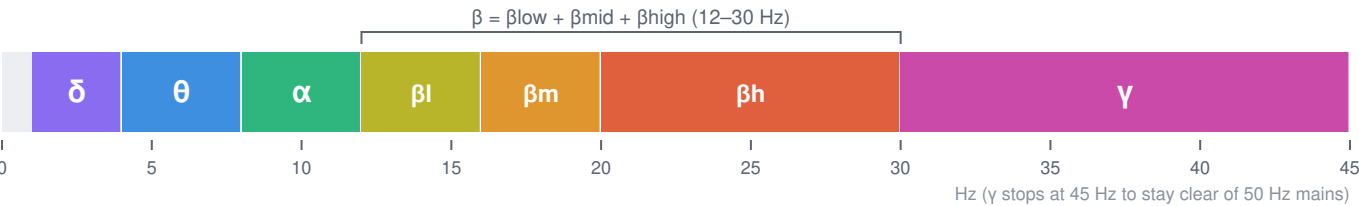


EEG band-power ratios

oak_bci · Unicorn Hybrid Black
8 channels averaged · 2026-10-05

What each relaxation metric measures, and how it drives the tree

1 THE FREQUENCY BANDS

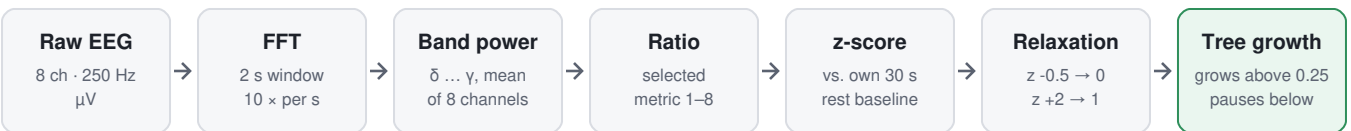


Δ Delta 1–4 Hz Deep sleep. Awake: mostly blink & movement artifact.	θ Theta 4–8 Hz Drowsiness, deep meditation, memory. Frontal θ = focus.	α Alpha 8–12 Hz Relaxed wakefulness; strongest with eyes closed, at the back.	β Beta 12–30 Hz Alert, busy mind, mental effort, muscle tension.	γ Gamma 30–45 Hz Cognitive binding; on scalp EEG mostly muscle artifact.
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2 THE EIGHT METRICS (keys 1–8 in the HUD)

#	RATIO	WHEN RELAXING	WHAT IT MEANS
1	α / β Alpha / Beta	RISES	Relaxed vs. engaged: alpha grows and beta falls as effort and tension drop. <i>Simple and intuitive; closing the eyes alone boosts alpha strongly.</i>
2	(α + θ) / β (Alpha + Theta) / Beta	RISES default	Relaxation deepening toward meditation: calm alpha and slow theta over beta. <i>Classic alpha–theta neurofeedback ratio; also rises if the user gets drowsy.</i>
3	α / total Relative Alpha	RISES	Share of all EEG power that is alpha; ignores overall amplitude (contact, gain). <i>Blinks and movement add delta to the total and pull it down.</i>
4	α / (θ + β) Alpha / (Theta + Beta)	RISES	Calm but awake: rewards alpha, penalises both tension (β) and drowsiness (θ). <i>Strictest score; use when falling asleep must not count as relaxing.</i>
5	(α + 0.5θ) / β Relaxation Score	RISES	Middle ground between 1 and 2: theta counts half, so drowsiness helps less. <i>The 0.5 weighting is a heuristic, not a published standard.</i>
6	θ at Fz Frontal Theta	RISES with focus	Frontal-midline theta: marker of sustained inward attention (focused meditation). <i>Absolute power at one electrode, not a ratio: sensitive to contact quality.</i>
7	θ / α Theta / Alpha	RISES when drowsy	Shift from relaxed wakefulness toward sleep onset or very deep meditation. <i>Cannot tell deep meditation from dozing off; best as a drowsiness monitor.</i>
8	θ / β Theta / Beta	MIXED	Classic attention-marker (high = low alert arousal), used in ADHD neurofeedback. <i>Not meditation-specific; mixes relaxation with inattention.</i>

3 FROM SIGNAL TO TREE



Smoothed over 1.5 s and blended over 0.75 s when switching metric, so the tree never jumps.

4 READING THE NUMBERS

- **Only relative values count.** Absolute ratios differ between people, sessions and electrode contact; that is why every metric is z-scored against the participant's own baseline.
- **Artifacts mimic states.** Eyes closed → α up. Blinks or head movement → δ up. Jaw clench or frown → β and γ up (looks like tension). Poor contact → flat line or 50 Hz hum.
- **Picking a metric.** Start with 2, (α+θ)/β. Switch to 4 if participants drift off to sleep, or to 6 for focused-attention meditation.